

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5

Duration :60 Mins

Total Marks:50

Assessment Mode: In Class

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Arunprasath R 192524077(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

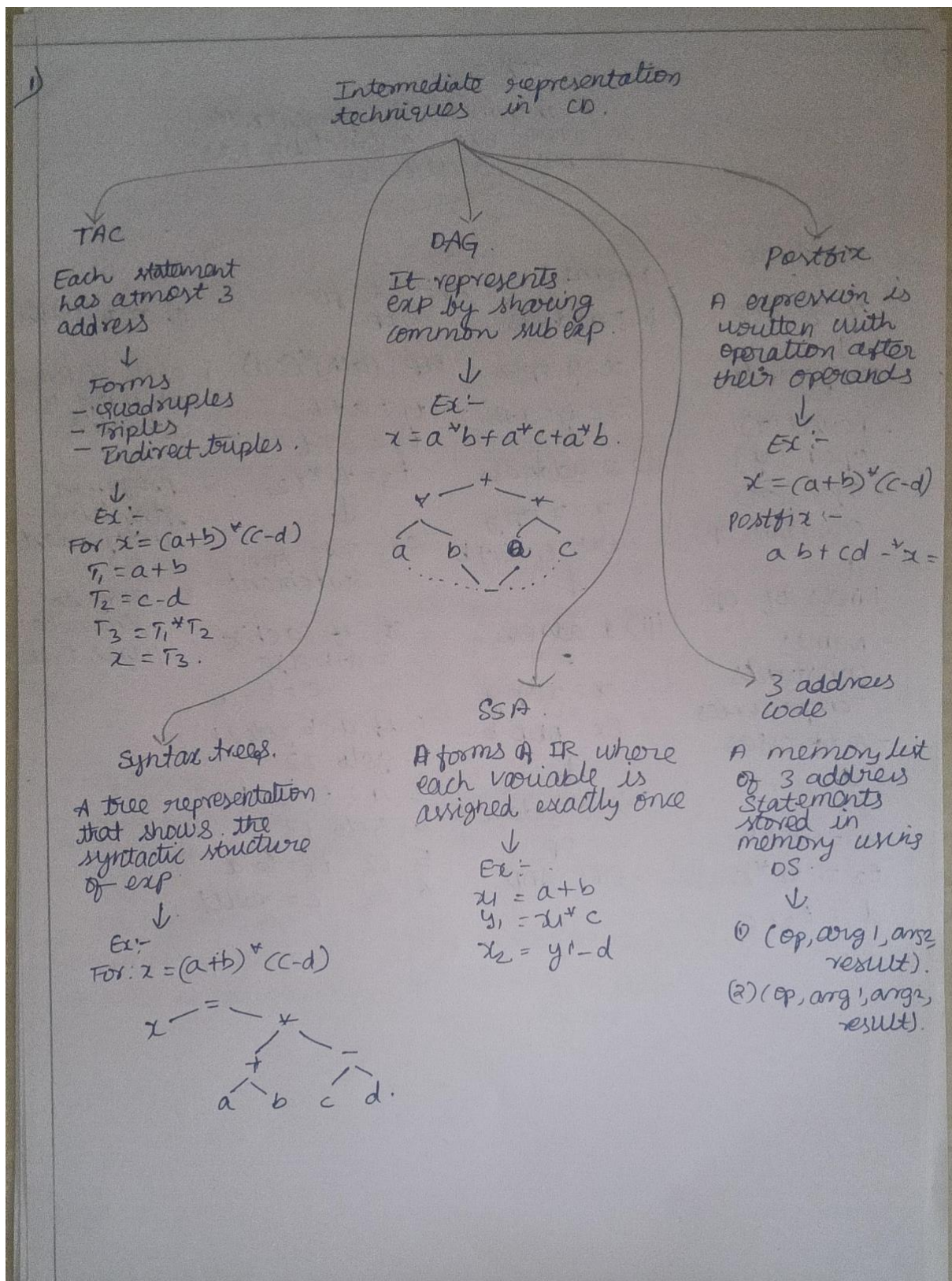
Arunprasath R

Q. No	Understanding of Data (25)	Analysis (25)	Application of Concepts (20)	Conclusion (20)	Clarity (10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 25	/ 20	/ 20	/ 10	/ 100

Annexure A

CONCEPT MAPPING

1. Intermediate Representation Techniques in Compiler Design

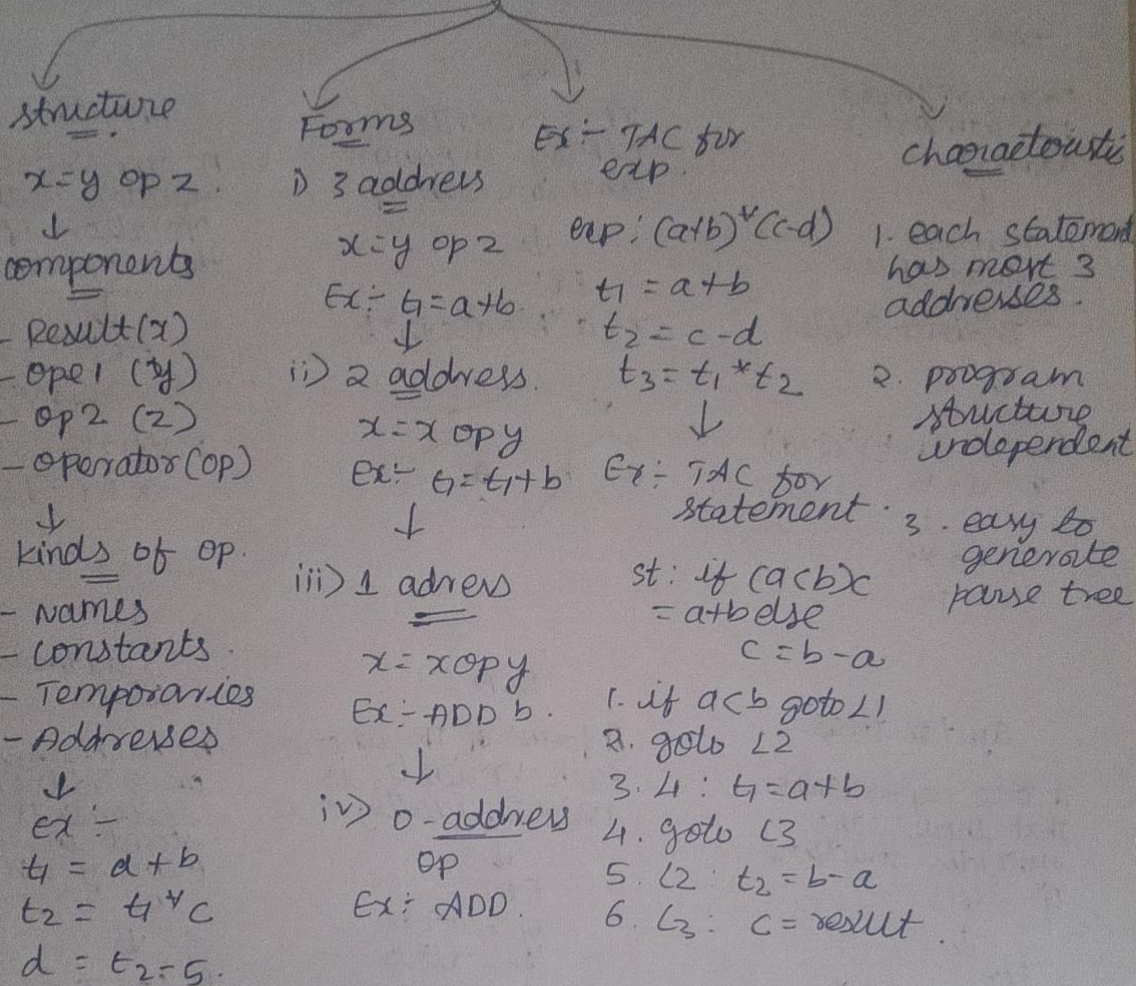


2. Structure and Characteristics of Three-Address Code

2)

TAC

A intermediate representation
in which each instruction has
at most 3 address.



3. Representation of TAC Using Quadruples, Triples and Indirect Triples

③

Rep of TAC using
Quadr, Triple, indirect tri

Quadruples

each instruction is
rep by records with
4 fields.

structure-

op	arg1	arg2	result
----	------	------	--------

Ex:-

$a = b + c * d - e$

op	arg1	arg2	r
(1) =	c	d	t1
(2) +	b	t1	t2
(3) -	t2	e	a

Advantage:

- clear representation
- Easy for code generatn.

characteristic

- simple & easy.
- uses more memory

Triples

Each inst is
rep by 3 fields.

str:-

op	arg1	arg2
----	------	------

Ex:-

$a = b + c * d - e$

op	arg1	arg2
(1) =	c	d
(2) +	b	(1)
(3) -	(2)	e

Advantage:

- Good for temporary values

characteristic

- only 3 field stored

Indirect

An array of
pointers to
tuples.

Example:

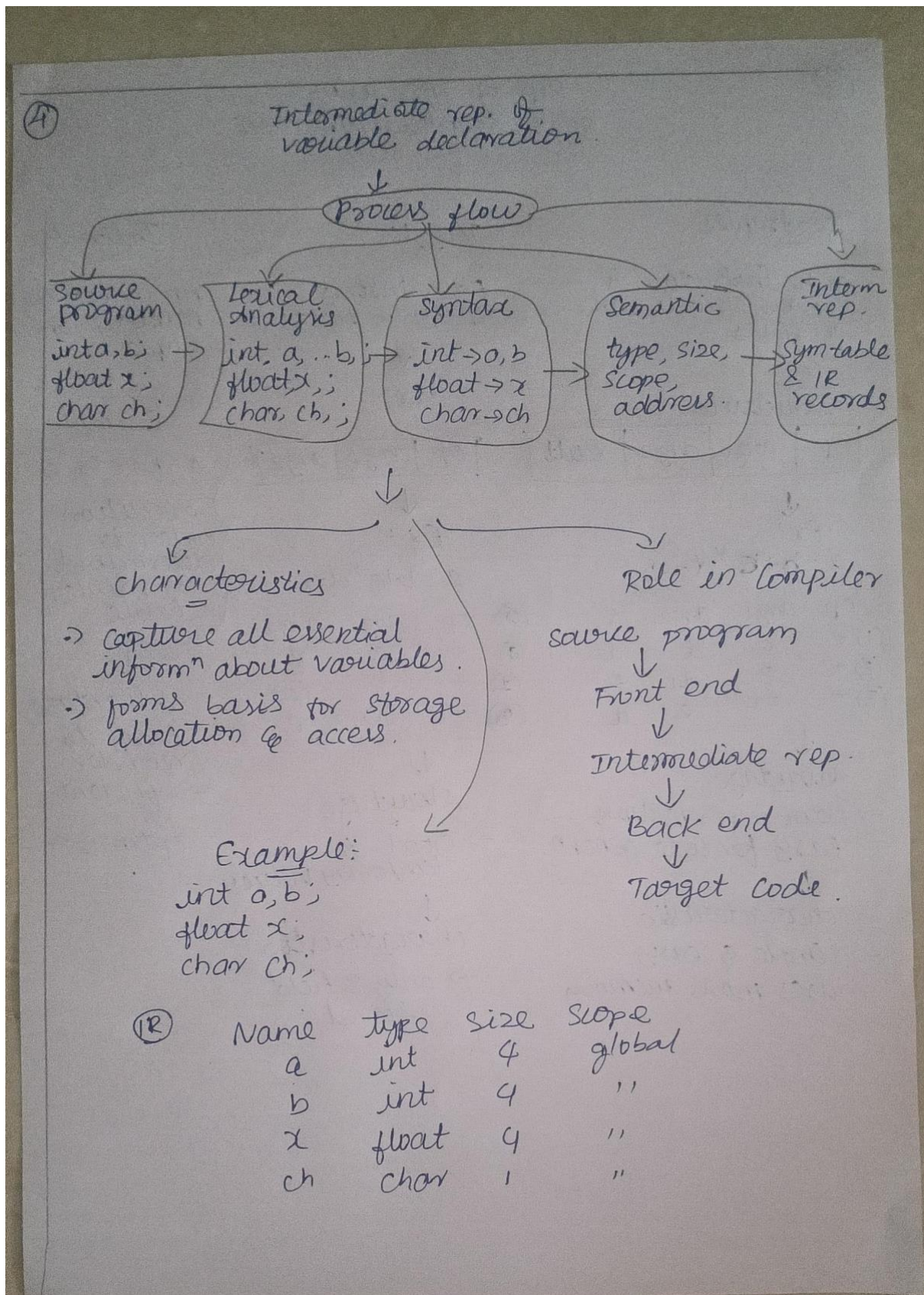
$a = b + c * d - e$

execution
order is
determined
by pointer
table

Advantage

- Easy to reorder
- efficient for optimizn

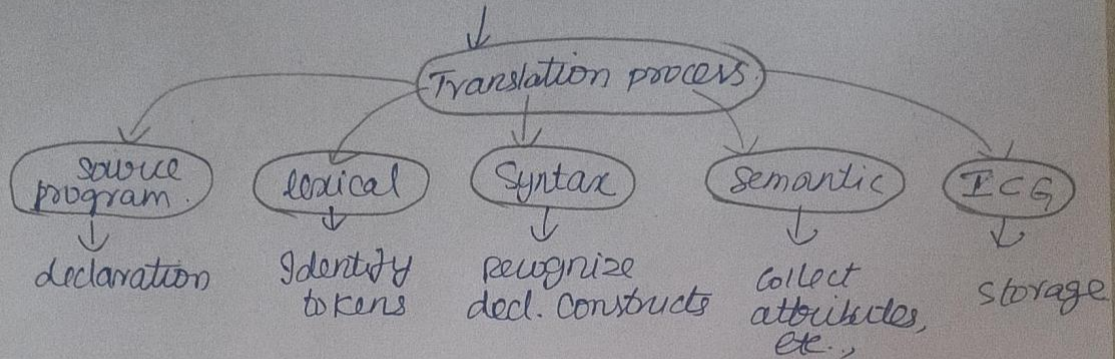
4. Intermediate Representation of Variable Declarations



5. Translation of Declarations into Intermediate Code

⑤

Translation of declaration
into I. code.



IR for declarations - forms

- storage Allocation (no initializⁿ)
alloc <size> for <name>
- storage Allocation (initializⁿ)
alloc <size> for <name>
<name> = <initial_value>
- Array.
alloc <size = n> for <name>

Examples →
 int a, b;
 float x = 3.5;
 char ch;
 int arr[10];